Meet 0
October 19, 2026
Team

The answer key is below.

p	Answer	p	Answer
1	67	4	1790
2	181	5	140
3	82	6	104190443

1. Suppose p is a prime such that $q = (p^2 - 1)^2 + 3$ is also a prime. Compute q.

   Proposed by: Ahmed Ilyasov

   Answer: $\boxed{67}$

   Solution: If $p \not\equiv 0 \pmod 3, p^2 \equiv 1 \pmod 3$, so $p^2 - 1 \equiv 0 \pmod 3$. Therefore,

   $$q = (p^2 - 1)^2 + 3 \equiv 0 \pmod 3.$$

   Since q is prime, this forces $q = 3$, but since $(p^2 - 1)^2 + 3 > 3$ for every prime p, we get a contradiction. Hence, $p \equiv 0 \pmod 3$, and since p is prime, $p = 3$. This yields $q = (3^2 - 1)^2 + 3 = 8^2 + 3 = \boxed{67}$.

2. Compute the sum of the numerator and denominator of the expression

   $$1 + \frac{1}{2 + \frac{1}{4 + \frac{1}{8}}}$$

   when fully simplified.

   Proposed by: Ahmed Ilyasov

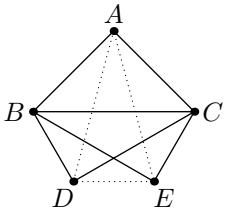
   Answer: $\boxed{181}$

   Solution: Note that

   $$1 + \frac{1}{2 + \frac{1}{4 + \frac{1}{8}}} = 1 + \frac{1}{2 + \frac{1}{\frac{33}{8}}} = 1 + \frac{1}{2 + \frac{8}{33}} = 1 + \frac{1}{\frac{74}{33}} = 1 + \frac{33}{74} = \frac{107}{74}.$$

   Now, $107 + 74 = \boxed{181}$.

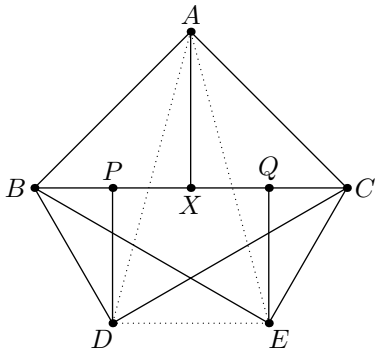
3. In $\triangle ABC$, $\overline{AB} = \overline{AC}$, $\overline{BC} = 10$ and $\angle A = 90^\circ$. Point D lies on the opposite side of $\overline{BC}$ as A, such that $\angle DBC = 60^\circ$ and $\angle DCB = 30^\circ$. Point E lies on the same side of $\overline{BC}$ as D such that $\angle EBC = 30^\circ$ and $\angle ECB = 60^\circ$. The area of $\triangle ADE$ is $\frac{a+b\sqrt{c}}{d}$ where a, b, c, and d are positive integers with $\gcd(a, b, d) = 1$ and c is not divisible by the square of any prime. Compute $a + b + c + d$.

   Proposed by: Ahmed Ilyasov

   Answer: $\boxed{82}$

   Solution:



Let P and Q be the altitudes from D and E, respectively to line $\overline{BC}$. Note that $\angle PBD = 60°$, so $\triangle BPD$ is another 30-60-90 triangle, meaning $\overline{BP} = \frac{1}{2}\overline{BD} = \frac{1}{4}\overline{BC} = 2.5$. Similarly, $\overline{CQ} = 2.5$. Now, since triangles $\triangle BCD$ and $\triangle CBE$ are congruent, we have their altitudes are congruent, so $\overline{DP} = \overline{EQ}$. However, we also know $\overline{DP} \parallel \overline{EQ}$, so $DPQE$ is a rectangle, meaning $\overline{DE} = \overline{PQ} = 5$.

We will now calculate the altitude from A to $\overline{DE}$. Let X be the midpoint of $\overline{BC}$. Note that $\overline{AX} \perp \overline{BC}$, and thus, $\overline{AX} = \frac{1}{2}\overline{BC} = 5$. Now, the distance from X to $\overline{DE}$ is the same as the distance from P to $\overline{DE}$, which is equal to $2.5\sqrt{3}$ using the 30-60-90 triangle. Thus, the length of the altitude from A to $\overline{DE}$ is equal to $5 + 2.5\sqrt{3}$.

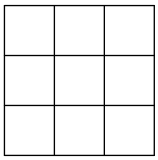
Thus, the area of $\triangle ADE$ is

$$5 \cdot \left(5 + \frac{5\sqrt{3}}{2}\right) \cdot \frac{1}{2} = \boxed{\frac{50 + 25\sqrt{3}}{4}},$$

so the answer is $50 + 25 + 3 + 4 = \boxed{82}$.

4. Place digits in a 3×3 grid such that

   - when the rows are read from left to right, the first row is a multiple of 67, the second row is $n^2 + 2n$ for an integer n, the third row is $a^b + b^a$ for integers $a, b > 1$,
   - when the columns are read from top to bottom, the first column is a perfect square, the second column is a Fibonacci number, and when the third column is multiplied by 7, the last three digits are 999,
   - and none of the rows or columns start with a 0.



Compute the sum of the three-digit numbers formed by the rows.

Proposed by: Taison Scofield

Answer: $\boxed{1790}$

Solution: Column 3 implies $7n \equiv -1 \pmod{1000}$. Since 143 is the multiplicative inverse of 7 (mod 1000), we get $n \equiv -143 \equiv 857 \pmod{1000}$, so Column 3 is 857.

The only two possibilities for Row 1 are $67 \cdot 4 = 268$ and $67 \cdot 14 = 938$. Suppose Row 1 is 268. Then Column 2 must be 610, so Row 2 ends in 15. From the rule for Row 2, notice

that $n^2 + 2n = (n+1)^2 - 1$. Thus, there must be a three-digit square ending in 16, which is impossible. Therefore, Row 1 must be 938.

Now Column 2 must be 377 and Column 1 must be 961, completing the grid. We can check that Row $2 = 26^2 - 1 = 675$ and Row $3 = 2^7 + 7^2 = 177$.

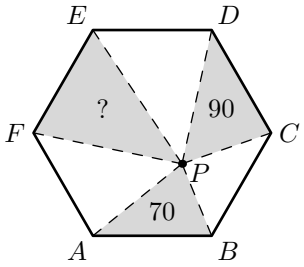
Our completed grid is

9	3	8
6	7	5
1	7	7

,

so the answer is $938 + 675 + 177 = \boxed{1790}$.

5. Regular hexagon $ABCDEF$ has area 600. Point P lies inside the hexagon so that $\triangle PAB$ has area 70 and $\triangle PCD$ has area 90. Find the area of $\triangle PEF$.



Proposed by: Steven Zheng

Answer: $\boxed{140}$

Solution: Let $\mathcal{T}$ be the triangle formed by the lines AB, CD, and EF with vertices X, Y, and Z. Then

$$[\mathcal{T}] = [PXY] + [PYZ] + [PZX] = 3([PAB] + [PCD] + [PEF]).$$

Now the area of the hexagon is $\frac{2}{3}$ that of $\mathcal{T}$, so $\mathcal{T}$ has area 900, so the sum of the area of the shaded triangles is 300. Hence the answer is $300 - 70 - 90 = \boxed{140}$.

6. Michael owns 10 000 dollars in Minnetonka Math Team stock. Each day, the stock randomly goes either up or down ten percent with equal probability. After four days, Michael's stock is worth d dollars. Compute the expected value of $(d+6)(d+7)$.

Proposed by: Michael Luo

Answer: $\boxed{104190443}$

Solution: Let X_i be the random variable corresponding to the value of Michael's stock after i days. We have

$$\mathbb{E}[X_{i+1}] = \frac{1}{2}(\mathbb{E}[0.9X_i] + \mathbb{E}[1.1X_i]) = \mathbb{E}[X_i]$$
$$\mathbb{E}[X_{i+1}^2] = \frac{1}{2}(\mathbb{E}[(0.9X_i)^2] + \mathbb{E}[(1.1X_i)^2]) = 1.01\mathbb{E}[X_i^2],$$

so we compute

$$\mathbb{E}[(X_4+6)(X_4+7)] = \mathbb{E}[X_4^2] + 13\mathbb{E}[X_4] + 42 = 10^8 \cdot 1.01^4 + 13 \cdot 10^4 + 42 = \boxed{104190443}.$$

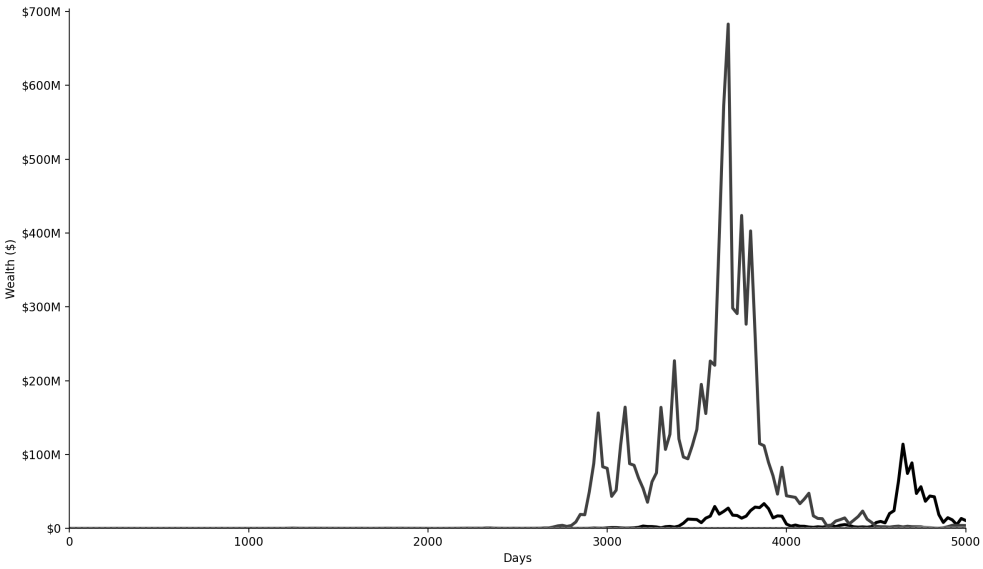


Figure 1: A simulation of 50000 portfolios for 5000 days. The top three "lottery winners" are shown and end with 10765014, 3946969, and 792615 dollars, respectively. Note the second lottery winner reaches a peak of almost 700 million dollars before falling down to under 4 million dollars. Only 10 of the 50000 make a profit, while 49381 end with less than 1 dollar. Figure generated with Codex.

Remark. This problem is somewhat unintuitive. In the long run, Michael goes bankrupt, which means for any $\varepsilon > 0$ the probability Michael has less than ε dollars goes to 1 as the number of days approaches infinity. Intuitively, this is since going up then down 10 percent leaves Michael with 1 percent less than he started with, so as long as Michael has approximately as many good and bad days, these 1 percents will add up and Michael will be left bankrupt. Equivalently, the logarithm of Michael's wealth is a random walk, increasing by $\log(1.1) \approx 0.095$ and decreasing by $-\log(0.9) \approx 0.105$ with equal probability. Therefore, the logarithm of Michael's wealth approaches $-\infty$ almost surely, hence Michael's wealth approaches 0 almost surely.

However, the expected value of Michael's wealth remains 10^4, as seen above. The interpretation is that Michael "wins the lottery" and achieves massive wealth with tiny probability, but goes bankrupt in most worlds. Most of the expected value comes from the times Michael wins the lottery (Figure 1).

The situation is even weirder when considering the square of Michael's wealth. Even though Michael's expected squared wealth grows exponentially in time here, the median run goes bankrupt twice as quickly as the previous case.